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Research Article

Zone 2 Intensity: A Critical Comparison of Individual Variability in Different Submaximal Exercise Intensity Boundaries

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Introduction: Endurance athletes often utilize low-intensity training, commonly defined as Zone 2 (Z2) within a five-zone intensity model, for its potential to enhance aerobic adaptations and metabolic efficiency. This study aimed at evaluating intra- and interindividual variability of commonly used Z2 intensity markers to assess their precision in reflecting physiological responses during training.
Methods: Fifty cyclists (30 males and 20 females) performed both an incremental ramp and a step test in a laboratory setting, during which the power output, heart rate, blood lactate, ventilation, and substrate utilization were measured.
Results: Analysis revealed substantial variability in Z2 markers, with the coefficients of variation (CV) ranging from 6% to 29% across different parameters. Ventilatory Threshold 1 (VT_1) and maximal fat oxidation (Fat_{Max}) showed strong alignment, whereas fixed percentages of HR_{max} and blood lactate thresholds exhibited wide individual differences.
Discussion: Standardized markers for Z2, such as fixed percentages of HR_{max}, offer practical simplicity but may inaccurately reflect metabolic responses, potentially affecting training outcomes. Given the considerable individual variability, particularly in markers with high CVs, personalized Z2 prescriptions based on physiological measurements such as VT_1 and Fat_{Max} may provide a more accurate approach for aligning training intensities with metabolic demands. This variability highlights the need for individualized low-intensity training prescriptions to optimize endurance adaptations in cyclists, accommodating differences in physiological profiles and improving training specificity.

Keywords: aerobic exercise; exercise metabolism; moderate intensity; substrate utilization; training prescription

1. Introduction

Endurance exercise elicits a broad spectrum of metabolic and morphological adaptations in skeletal muscle (such as increased capillarization, elevated activity of enzymes involved in the mitochondrial electron transport chain, and decreased carbohydrate utilization), designed to attenuate cellular perturbations during subsequent training sessions [1]. These chronic adaptations are hypothesized to result from the cumulative effects of repeated exercise bouts, with

initial signaling events triggering the adaptive processes following each session [1]. For practical application and to optimize the targeting of specific adaptations, exercise intensity is systematically categorized into distinct training zones [2], such as Zone 2, threshold [3], or SIT [4]. These zones are typically derived from various testing procedures, including ramp or incremental step testing protocols [5–8].

Depending on the sport and discipline, multiple intensity models exist, with the most commonly used frameworks being the three-zone, five-zone, and seven-zone models [2, 6, 9, 10]. These models provide a structured framework for training intensity prescription, allowing for the modulation of exercise intensity to selectively engage various energy systems (i.e., predominantly phosphagenic, glycolytic, or oxidative), muscle fiber types, and cardiovascular mechanisms [1, 11, 12]. The choice of model may vary depending on the sport and federation (e.g., Norway [13] and Germany [14]), with each framework offering a different level of granularity in intensity prescription. For instance, the three-zone model divides intensity into broad categories suitable for monitoring overall training load, while the five- and seven-zone models enable a more detailed approach, with additional zones designed to target specific physiological adaptations (e.g., neuromuscular or cardiac adaptations and upregulation of mitochondrial enzymes [15, 16]).

Although no standardized criteria currently exist for defining these training zones [6], each zone is often determined by objective physiological markers, such as heart rate and blood lactate concentration [17–28] and ventilatory response [9, 29–32], or by power output or velocity [24, 33–40]. Based on the athlete's specific needs and objectives, this zonal approach enables the individualized tailoring of training stimuli to elicit targeted metabolic and performance outcomes, such as enhancing aerobic capacity, optimizing fuel utilization [41], or improving lactate clearance efficiency [2, 42, 43].

Training in intensity Zone 2 (Z2), based on a five-zone intensity model, is particularly popular in endurance sports, especially those with decreased mechanical stress (such as cycling or cross-country skiing), due to its role in enhancing aerobic capacity, efficiency, and metabolic flexibility [44]. Z2 training is positioned in the low-intensity range, making it a foundational component of training for both elite and amateur athletes [2, 44]. The primary rationale behind extensive Z2 exercise lies in its ability to stimulate mitochondrial efficiency and enhance fat metabolism [41, 44, 45] among a variety of adaptations (e.g., insulin sensitivity, oxidative capacity, glucose control, and increased glucose stores) [46, 47]. This supports sustained energy production during long-duration efforts by preserving glycogen stores and delaying fatigue, enabling athletes to maintain high performance over extended periods—a key factor for both professional and amateur athletes [48]. While not a new concept [49], elite cyclists, including recent Tour de France champions, frequently incorporate Z2 training [50], dedicating approximately 80% of their training time to this intensity and below [51].

With the widespread availability of power meters and heart rate sensors, amateur cyclists often rely on accessible, nonlaboratory-based markers to guide their training [3]. These markers include $\%HR_{max}$, heart rate reserve (HRR), or percentages of peak power output (%PPO), which are used to target physiological goals such as maximizing fat oxidation (Fat_{Max}) and staying within an "aerobic zone." Z2 intensity typically corresponds to approximately 72%–82% of HR_{max} [10], 1.5–2.5 mmol/L blood lactate [10], or 60%–75% of functional threshold power (FTP) [3] in cycling.

Fat_{Max}, in particular, has become a popular objective within Z2 training due to its role in improving endurance performance and extending time to fatigue during prolonged events [52]. To further define Z2 intensity, additional physiological indicators are employed, including Ventilatory Threshold 1 (VT_1) [9], 65% $\dot{V}O_{2peak}$ [53], and blood lactate concentrations within the range of 1.5–2.5 mmol/L ($BLa_{1.5}$, $BLa_{2.0}$, $BLa_{2.5}$) [10], to further help distinguish intensity boundaries.

However, despite the widespread application of Z2 training, there is growing evidence that commonly used physiological markers—such as % of HR_{max} or metabolic reference points such as capillary blood lactate thresholds (e.g., power output at 2 mmol/L blood lactate, P_2)—may not result in consistent metabolic responses across individuals [54, 55]. This variability arises not only between individuals with different aerobic capacities but even within subgroups of athletes with similar fitness levels [55].

Such findings suggest that the conventional approach of using fixed percentages or numbers to define the upper and lower boundaries of Z2 may not ensure the desired metabolic perturbation across all athletes. Consequently, using generalized percentages or surrogate markers may result in exercise prescriptions that are either too low or too high for specific individuals, which could undermine the effectiveness of training.

To ensure a training intensity domain such as Z2 elicits the targeted metabolic load, it is essential to account for the significant interindividual variability in physiological responses. This study aimed at analyzing the extent of variability in the most commonly applied reference values for Z2 as this variability is critical for determining the precision and effectiveness of Z2 training. Specifically, we aim to determine whether the current methods for prescribing Z2 training—based on the heart rate, blood lactate, ventilatory indices, or power—consistently lead to similar exercise intensities across individuals, or if a more individualized approach is required. We hypothesized that non-physiological Z2 demarcations based on percentages of a maximal value display a large intraindividual variation, complicating their application as Z2 training prescriptors.

By investigating this variability, we aim to provide insights into the limitations of percentage- and number-based training methods in Z2, emphasizing the need for refined approaches in prescribing low-intensity exercise to optimize endurance training for cyclists.

2. Methods

2.1. Participants. A cohort of $n = 50$ ($n = 30$ males and $n = 20$ females) experienced cyclists with more than three years of regular cycling exercise (> 2 sessions per week) were recruited for this study [56]. All participants were

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experienced in road cycling with clipless pedals and cycled regularly as exercise. Prior to the study, the participants were informed of the protocol and gave their written informed consent to participate. All procedures were approved by the ethical committee of Exercise Science & Training of the Faculty of Human Sciences of the University of Würzburg (EV2024/1-1004) and conducted in accordance with the Declaration of Helsinki [57, 58]. The main characteristics of the participants are summarized in Table 1.

2.2. Experimental Design. Two experimental visits (T1 and T2) to the laboratory were required, which were at least 48 h apart and completed within a 7-day period to complete a ramp and a step incremental test. The overall study design is illustrated in Figure 1.

All participants were instructed to keep a nutrition diary and to repeat their usual diet for each visit within the 24 h before each experimental visit [59]. In addition, all were instructed to stay adequately hydrated, to eat a carbohydrate-rich meal (i.e., a banana and a jam sandwich) no less than 3 h before each visit, and to refrain from caffeine consumption on the day of testing. Each participant received 35 g of a carbohydrate mixture (IsoFast, DextroEnergy, Krefeld, Germany) dissolved in 500 mL of water to drink *ad libitum* during warmup and recovery periods.

2.3. Ramp Protocol. To determine $\dot{V}O_{2peak}$ and PPO_{ramp}, all participants performed a ramp protocol in T1. The participants began cycling at 100 W for 2 min with a freely chosen cadence, after which the load increased by 25 W every 30 s [60]. The ramp ended when volitional exhaustion was reached, or the cadence dropped by more than 10 rpm. Following 3-min active recovery at 75 W, a verification phase at 100% of the load at the last fully completed 30 s until volitional exhaustion was performed [61]. $\dot{V}O_{2peak}$ was calculated as the highest value averaged over 30 s.

2.4. Incremental Step Protocol. To determine power output at baseline blood lactate concentration +0.5 mmol/L ($BLa_{min+0.5}$), $BLa_{1.5}$, $BLa_{2.0}$, $BLa_{2.5}$, VT_1, 65% $\dot{V}O_{2peak}$, Fat_{Max}, $HR_{72\%}$, and $HR_{82\%}$, a step protocol was performed (T2). The participants began the test with a freely chosen cadence but were advised to maintain their regular cadence. The test started at 100 W, with the load increasing by 25 W every 3 min. Three participants wished to start at 75 W based on previous results. The test concluded when volitional exhaustion was reached or when cadence consistently dropped by more than 10 rpm. Peak power output (PPO) during the step test was estimated using the following formula:

$$PPO_{step} = P_{\text{last fully completed stage}} + \frac{t_{\text{in last stage}}}{180} * 25. \quad (1)$$

2.5. Cycling Ergometer. All cyclists used their own shoes and pedals for both tests, and the tests were performed on the cyclist's own personal road bike installed on a Cyclus2 ergometer (RBM, Leipzig, Germany). The Cyclus2 is an electromagnetically braked ergometer and measures power with an accuracy error of 2% according to the manufacturer. For both visits, all cyclists warmed up for 10 min cycling at 1.5 W/kg body mass, then performed a short sprint before undergoing an incremental ramp (T1) or step (T2) protocol.

2.6. Gas Exchange and Heart Rate Measurement. Participants were fitted with a Hans Rudolph V2 mask (Hans Rudolph, Inc, Shawnee, KS, USA), and expired gases along with breathing volume were analyzed using a Cosmed Quark CPET system (Cosmed Srl, Rome, Italy). Gas and volume analyzers were calibrated before each test using precision gas (16% O_2 and 5% CO_2) and a volume pump, following the manufacturer's instructions (Airgas Therapeutics, Plumsteadville, PA, USA). Gas exchange parameters were averaged over 10-s intervals, with only the final 30 s of each step being considered for analysis. VT_1 was determined following the method described in Ref. [62] and assessed by two independent experienced researchers in exercise testing and prescription who were blinded to the identity of the athletes and the aim of the study. Power output at VT_1 was linearly interpolated based on the test duration.

All were equipped with a Polar H10 (Polar Electro Oy, Kempele, Finland) heart rate belt. The heart rate minimum was the lowest value determined during the resting period before the warmup and thereafter. Maximum heart rate was determined as the maximal value averaged over 10 s.

2.7. Capillary Blood Lactate Concentration. Capillary blood samples of the left earlobe were sampled before the start of the step test and in the last 15 s of every step. The lactate concentration was measured amperometric-enzymatically employing Biosen C-Line (EKF Diagnostics, Barleben, Germany).

2.8. Z2 Definition and Determination of Thresholds. The following definitions were analyzed as representative benchmarks for defining "Zone 2" intensity:

- 72%–82% HR_{max} ($HR_{72\%}$, $HR_{82\%}$) and 1.5–2.5 mmol/L blood lactate used by the Norwegian Olympic Federation ($BLa_{1.5}$, $BLa_{2.5}$) [10].
- 65% of $\dot{V}O_{2peak}$ [53, 63].
- Blood lactate concentration of 2 mmol/L ($BLa_{2.0}$) [9, 17, 64].
- Baseline blood lactate concentration + 0.5 mmol/L ($BLa_{min+0.5}$) [65].
- First ventilatory threshold (VT_1) [9, 66, 67].
- Maximum fat oxidation rate (Fat_{Max}) [41, 45]. For the determination of Fat_{Max}, standard nonprotein stoichiometric equations for moderate intensity were applied according to Jeukendrup and Wallis [68],

$$\text{Fat oxidation}\left[\frac{g}{\text{min}}\right] = \left(1.695 * VO_2\right) - \left(1.701 * VCO_2\right). \quad (2)$$

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Table 1: Anthropometric data of participants.

Variable	All ($n = 50$)	Male ($n = 30$)	Female ($n = 20$)
Age (years)	31.2 ± 7.8	34.3 ± 7.8	26.5 ± 5.0
Height (cm)	177.5 ± 9.3	182.6 ± 7.0	169.8 ± 6.6
Body mass (kg)	71.7 ± 12.4	78.6 ± 10.3	61.4 ± 6.6
Body fat (%)	13.9 ± 4.8	11.7 ± 3.8	17.1 ± 4.2
Relative oxygen uptake (mL/min/kg)	54.1 ± 8.1	56.0 ± 8.6	51.2 ± 6.6

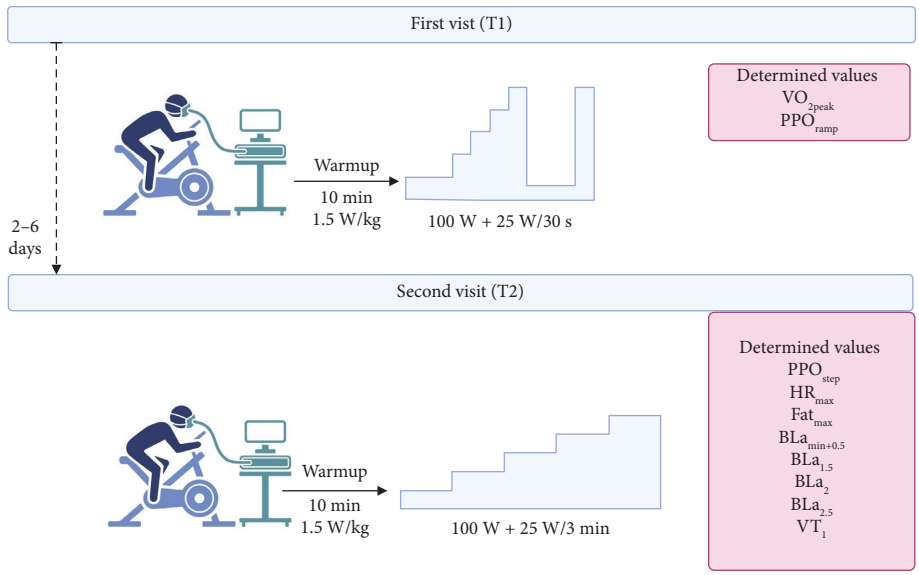

Figure 1: Study procedure.

2.9. Data Processing. All data were collected and exported in Microsoft Excel. All statistical analyses were performed in GraphPad Prism (v.10.4, Boston, MA, USA).

HR, $\dot{V}O_2$, fat oxidation, and blood lactate were matched to the corresponding power outputs at the end of each step. Slopes and intercepts of HR and $\dot{V}O_2$ were aligned with power output via individual linear regressions, based on values measured at rest, at the end of each stage, and at maximum effort. All linear regressions for HR and $\dot{V}O_2$ achieved a goodness of fit ($R^2 \geq 0.95$). Lactate values were interpolated using third-order polynomial regression analysis, with all regressions attaining goodness of fit ($R^2 \geq 0.97$).

For all blood lactate-based thresholds ($BLa_{1.5}$, BLa_2, $BLa_{2.5}$, and $La_{min+0.5}$), power at the corresponding lactate level was calculated using third-order polynomial regression, with matching HR and $\dot{V}O_2$ determined via linear regression. Nine participants did not reach a lactate value lower than 1.5 mmol/L and were excluded from this analysis. For VT_1, power output at VT_1 was interpolated, and matching HR and $\dot{V}O_2$ were similarly determined via linear regression. For 65% of $\dot{V}O_{2max}$, power output was calculated and matched to HR via linear regression. For $HR_{72\%}$ and $HR_{82\%}$, the respective percentages of HR_{max} were calculated and matched to power output and HR using linear regression. For Fat_{Max}, the power output corresponding to the highest total fat oxidation in the final 30 s was determined, with matching HR and $\dot{V}O_2$ calculated via linear regression [69].

2.10. Statistical Analysis. All raw data were collected using Microsoft Excel (Microsoft Corp., Redmond, WA, USA). Statistical analyses (mean, standard deviations, and 95% confidence intervals) were computed with GraphPad Prism (v10.3, Boston, MA, USA). Normality of the power output, heart rate, and $\dot{V}O_2$ was assessed using the Shapiro–Wilk test, without requiring further transformation, except for power output at Fat_{Max} and corresponding $\dot{V}O_2$. Median and mean absolute deviation (MAD) are therefore given for this demarcation [70], calculated with JASP (v0.18.3). The level of significance (α) was set to 0.05 for all statistical analyses.

Differences between the position of markers and the respective maximum values were analyzed employing a paired mixed-effects model because of missing values in one Z2 demarcation. Tukey's multiple comparison test was applied to identify differences between the different Z2 demarcations.

The interindividual coefficient of variation (CV) was calculated as the standard deviation divided by the mean for absolute values of power, heart rate, and oxygen uptake of the given Z2 demarcations [71]. For Fat_{Max}, a robust version of CV (based on robust MAD) was calculated according to Arachchige et al. [72]. Bland–Altman plots were used to assess agreement, with bias and limits of agreement calculated as the difference between Z2 demarcations and their average [73].

Differences between males and females were analyzed by independent sample Student's t-tests.

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3. Results

The mean values, standard deviations, and CV for each variable reflecting Z2 in relation to the peak oxygen uptake, peak heart rate, and PPO are summarized in Table 2. The CVs ranged from 6% to 29% with $HR_{72\%}$, $HR_{82\%}$, and $BLa_{2.5}$ displaying the lowest variation, while power output at $BLa_{1.5}$ exhibited the highest CV.

The rain cloud plots in Figure 2 show single data points for each individual and variable related to Z2 in relation to $\dot{V}O_{2peak}$, HR_{max}, HRR, PPO_{ramp}, and PPO_{step}.

Upon analyzing the absolute power output values across different Z2 demarcations, the multiple comparison test revealed no significant differences between the combinations of $BLa_{min+0.5}$, $BLa_{1.5}$, VT_1, and $HR_{82\%}$, as well as between $HR_{72\%}$ and Fat_{Max} ($p > 0.05$). However, all other pairwise comparisons demonstrated significant differences ($p < 0.01$). Figures 3(a) and 3(b) display the mean differences and 95% confidence intervals with VT_1 and Fat_{Max} as reference markers.

When related to % of PPO_{ramp} and PPO_{step}, there was similar overlay (i.e., no significant differences, $p < 0.05$) for $BLa_{min+0.5}$ and $BLa_{1.5}$; $BLa_{min+0.5}$ and VT_1; $BLa_{min+0.5}$ and $HR_{82\%}$; $BLa_{1.5}$ and VT_1; $BLa_{1.5}$ and $HR_{82\%}$; $BLa_{1.5}$ and 65% $\dot{V}O_{2peak}$; BLa_2 and $HR_{82\%}$; $HR_{72\%}$ and Fat_{Max}; and VT_1 and $HR_{82\%}$. In contrast, all other comparisons displayed significant differences ($p < 0.01$).

When expressed as % of peak oxygen uptake, the mean values of $BLa_{min+0.5}$ and $BLa_{1.5}$; $BLa_{min+0.5}$ and VT_1; $BLa_{min+0.5}$ and $HR_{82\%}$; $BLa_{1.5}$ and VT_1; $BLa_{1.5}$ and $HR_{82\%}$; BLa_2 and $HR_{82\%}$; $HR_{72\%}$ and Fat_{Max}; and VT_1 and $HR_{82\%}$ did not differ ($p > 0.05$). In contrast, all other combinations showed significant differences ($p > 0.01$).

When related to % HR_{max}, no differences ($p > 0.01$) were found for $BLa_{min+0.5}$ and $BLa_{1.5}$; $BLa_{min+0.5}$ and BLa_2; $BLa_{min+0.5}$ and VT_1; $BLa_{1.5}$ and VT_1; $BLa_{1.5}$ and 65% $\dot{V}O_{2peak}$; BLa_2 and $BLa_{2.5}$; and BLa_2 and VT_1.

To illustrate the interconnection between physiologically determined Z2 demarcations and standardized ones, Figure 4 illustrates bias and limits of agreement of VT_1 and $BLa_{min+0.5}$; VT_1 and $HR_{82\%}$; and Fat_{Max} and $HR_{72\%}$.

Differences between males and females were found for power output at $BLa_{min+0.5}$ ($p = 0.0023$), $HR_{72\%}$ ($p = 0.0197$), $HR_{82\%}$ ($p = 0.0105$), BLa_2 ($p = 0.0116$), and $BLa_{2.5}$ ($p = 0.0167$) in relation to PPO_{step} and for heart rate at Fat_{Max} in relation to HR_{max} ($p = 0.0188$). Females displayed a higher ratio of their heart rate at Fat_{Max} compared to the maximal value than males. In all ratios of the power output, females displayed a lower ratio to PPO_{step}.

4. Discussion

The primary finding of this study is the significant interindividual and methodological variability observed in variables defining Z2 boundaries, including the heart rate, capillary blood lactate concentration, oxygen uptake, and power output. All these variables, commonly employed to characterize Z2, displayed notable individual differences. Based on our hypothesis, we thereby infer that Z2 demarcations not based on physiological measures are potentially suitable as proxy measures for Z2. However, the large intraindividual variability leads us to discourage from their general application in place of physiologically based diagnostic procedures.

From the mean statistical analysis, $BLa_{min+0.5}$, VT_1, and $HR_{82\%}$, as well as Fat_{Max} and $HR_{72\%}$, yielded comparable values. $BLa_{1.5}$ was also comparable to $BLa_{min+0.5}$, VT_1, and $HR_{82\%}$; however, only 41 of the 50 participants achieved a power output corresponding to this demarcation.

The present study reveals substantial variability in the heart rate and power output thresholds associated with Z2, underscoring the challenges in establishing standardized training prescriptions. Specifically, Z2 thresholds for the heart rate and power output exhibited CV exceeding 20%, which emphasizes the extent of interindividual differences in physiological responses, even among individuals with comparable aerobic capacities. These findings are consistent with prior research indicating heterogeneous metabolic responses at similar percentage-based thresholds (e.g., % $\dot{V}O_{2peak}$ and FTP%), highlighting the limitations of universal Z2 guidelines for precision in low-intensity endurance training [54, 55]. Given this inherent variability in defining Z2, focusing on variables with relatively low CV may provide a more consistent basis for comparison across individuals or groups, thereby enhancing the validity of Z2-based training prescriptions.

This variability likely originates from several individual physiological factors that influence energy production and utilization during exercise. The key aspects include the following:

i. Mitochondrial density [74], which determines the cell's capacity for oxidative metabolism and affects aerobic performance [1].
ii. Lactate kinetics, which dictates the rate at which lactate is produced, utilized, and cleared from the bloodstream.
iii. Oxygen transport efficiency, encompassing the cardiovascular and respiratory systems' ability to deliver oxygen to active muscles [75].
iv. Muscle fiber composition, particularly the ratio of slow-twitch to fast-twitch fibers, which impacts the preferred metabolic pathways during exercise [76, 77].

Together, these elements shape how blood lactate is produced, distributed, and cleared in response to various exercise intensities [78]. Since these factors vary significantly between individuals, they play a central role in determining the physiological response at submaximal intensities, thus impacting the accuracy and utility of training zones when prescribed based on fixed percentage guidelines [55]. As a result, it is unlikely that all athletes will achieve optimal adaptations from Z2 training prescribed solely by a fixed percentage of heart rate (e.g., 72%–82% of HR_{max}), as some may overestimate their intended intensity, while others may underestimate it [79, 80]. Although heart rate-based measures generally exhibit less interindividual variation,

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Table 2: Oxygen uptake, heart rate, and power output as mean ± SD, % of maximal respective value, and coefficient of interindividual variation for different Zone 2 demarcations.

	Oxygen uptake (mL/min) at	% peak oxygen uptake	CV%	HR (b/min)	% of HR_{max}	CV%	Power output (W)	% of PPO_{ramp}	CV%
$BLa_{min+}0.5$	2677 ± 576	69.4 ± 6.7	21.5	146 ± 10	81.2 ± 3.0	7.0	199 ± 45	51.2 ± 4.8	22.8
$BLa_{1.5}$	2654 ± 680	65.1 ± 13.2	25.6	141 ± 10	78.1 ± 6.9	7.6	194 ± 56	46.0 ± 13.6	29.0
$BLa_{2.0}$	2803 ± 661	72.5 ± 8.3	23.6	150 ± 9	83.4 ± 3.3	6.2	211 ± 53	54.1 ± 6.7	24.9
$BLa_{2.5}$	2951 ± 661	76.4 ± 7.7	22.4	154 ± 9	86.0 ± 3.0	6.0	224 ± 52	57.8 ± 6.1	23.3
VT_1	2690 ± 612	69.8 ± 8.9	22.7	146 ± 11	81.4 ±0 .3.9	7.3	200 ± 45	51.5 ± 5.9	22.8
65% VO_{2peak}	2513 ± 546	—	21.7	140 ± 15	78.2 ± 6.0	10.8	183 ± 49	47.1 ± 6.5	26.0
Fat_{Max}	2155 ± 407	56.0 ± 10.7	18.8	130 ± 12	72.3 ± 5.7	9.5	125 ± 37	38.7 ± 8.9	29.6
$HR_{72\%}$	2128 ± 525	55.2 ± 8.8	24.7	—	—	—	148 ± 41	38.0 ± 6.7	27.6
$HR_{82\%}$	2710 ± 592	70.3 ± 8.2	21.9	—	—	—	202 ± 46	51.9 ± 5.6	22.7
Maximum value	3866 ± 840	— —	21.7	179 ± 10		5.7	387 ± 73 (Ramp) 300 ± 56 (Step)	— —	18.9 18.7

a seemingly minor difference of 5% can translate to approximately 10 bpm, potentially posing a substantial difference in training intensity for the athlete.

Amateur cyclists often rely on accessible, nonlaboratory-based markers such as $\%HR_{max}$, HRR, or %PPO as references to target physiological goals such as Fat_{Max} or other metrics referring to an "aerobic zone," defined, e.g., by low blood lactate concentrations [3]. However, Figures 2(a), 2(b), 2(c), 2(d), and 2(e) illustrate that using such simplified intensity prescriptions can lead to highly heterogeneous physiological responses. In fact, we observed significant interindividual variability in Z2 intensity markers, such as $HR_{72\%}$–$HR_{82\%}$, a widely used definition within the five-zone model [10]. Acknowledging this variability, relying on fixed percentages of HR_{max} or blood lactate concentration for prescribing Z2 training may not yield similar anticipated results for all athletes.

Research indicates that males and females exhibit distinct physiological characteristics, particularly in their ratios of lactate threshold 1 (LT1) to PPO [81]. Our analysis also displayed differences between males and females in some cases. However, the differences we found were the opposite of those mentioned by Benítez-Muñoz [81]. This is possibly caused by methodological differences in both studies. Nonetheless, these findings underscore that there is a discrepancy between males and females and employing standardized percentages may not be equally suitable across the two groups.

The differences between males and females can be attributed to factors such as muscle composition, hormonal influences, and metabolic capacities [81]. For instance, women typically have a higher proportion of type I muscle fibers and greater capillary density, which enhance oxidative metabolism and endurance performance [82]. Hormonal variations, including differences in estrogen and progesterone levels, also play a role in substrate utilization and muscle function [82]. When training zones are determined using a standardized percentage of PPO, these inherent sex differences mean that the prescribed training intensities may not align equitably for both groups. Consequently, males and females could experience different metabolic responses to the same training stimulus, impacting the effectiveness of their training regimens. This discrepancy underscores the need for personalized training programs that consider individual physiological profiles. This ensures that each athlete receives the appropriate metabolic stimuli necessary to maximize performance and achieve training goals.

VT_1 marks the transition from predominantly aerobic metabolism to increased anaerobic contribution, aligning with a shift from fat to carbohydrate as the primary fuel source [41, 83]. In this case, using VT_1 to determine the upper boundary of the low-intensity exercise domain, including Z2, compared to relying on $HR_{72\%}$–$HR_{82\%}$ may represent a more individualized approach to ensure athletes train within an intensity range that optimally stimulates fat oxidation and promotes glycogen sparing [55, 84] while also facilitating intensity control [13].

The present data indicate that power at VT_1 and Fat_{Max} may overlap in some cyclists; however, Table 2 shows that Fat_{Max} is generally ~25% lower than VT_1—indicating that these markers represent distinct intensity domains [85, 86]. Fat_{Max} is often the rationale for many athletes to exercise at this intensity due to its association with optimal fat oxidation [45], which is particularly relevant for enhancing metabolic flexibility, i.e., the ability to transition between fats and carbohydrates as fuel sources depending on the availability and metabolic demand [41], improving endurance, and supporting body composition goals [87, 88]. For endurance athletes, training at Fat_{Max} can contribute to improved energy efficiency, allowing for the conservation of glycogen stores during prolonged exercise [45]. Additionally, exercising at this intensity may appeal to recreational athletes focused on weight management, as it targets fat as a major fuel source [87, 88].

Therefore, if Fat_{Max} is the desired physiological metric, we recommend directly assessing Fat_{Max} to optimize fat oxidation during exercise, rather than relying on surrogate intensity markers. It is also noteworthy that while training at Fat_{Max} may not elicit superior adaptations compared to other intensities [89], it provides an intensity with a high workload-to-carbohydrate utilization ratio, which is advantageous for those aiming to maximize fat utilization

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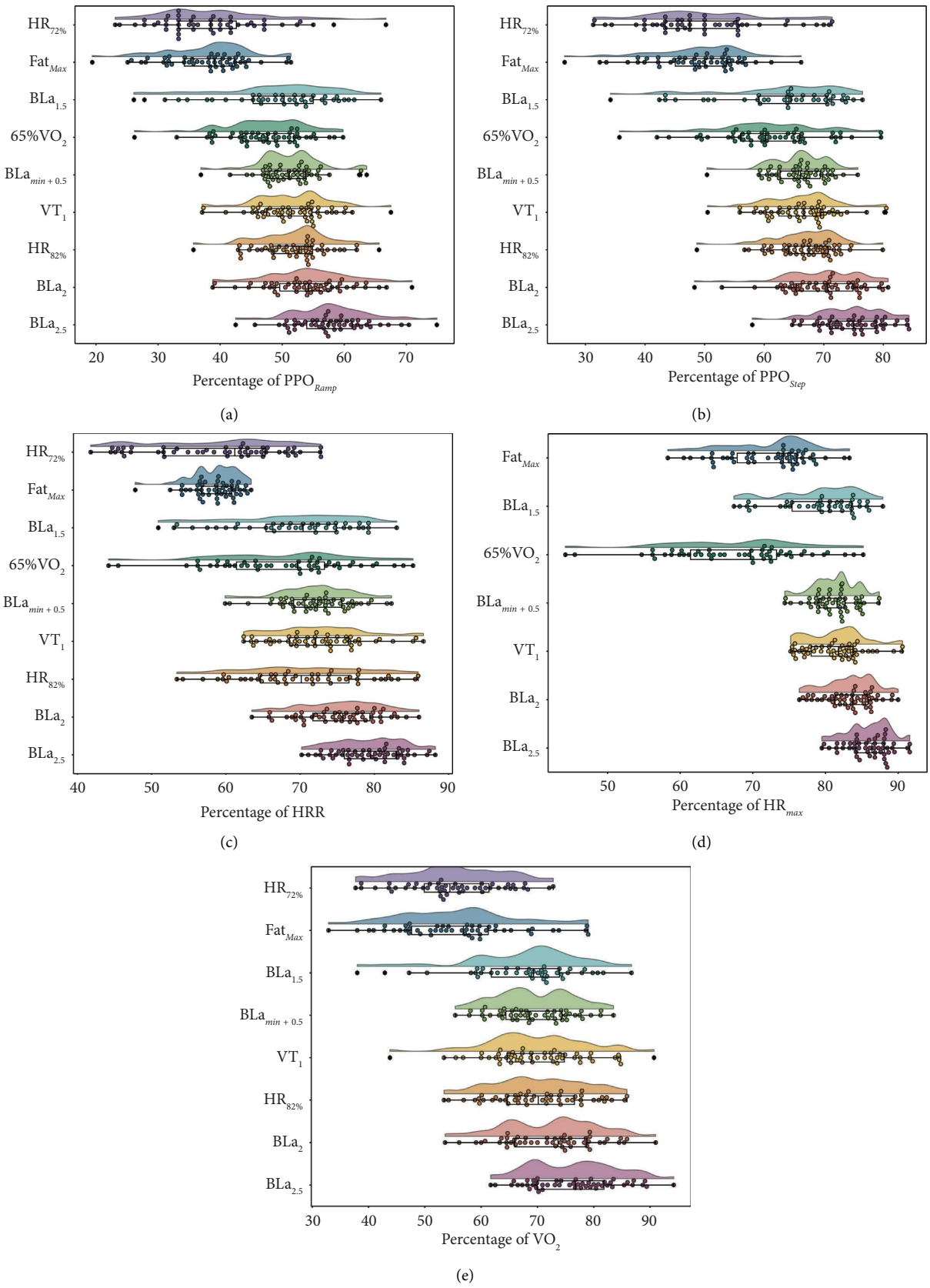

Figure 2: Raincloud plots illustrating Zone 2 demarcations expressed as power output, heart rate, or oxygen uptake relative to their respective maximum values.

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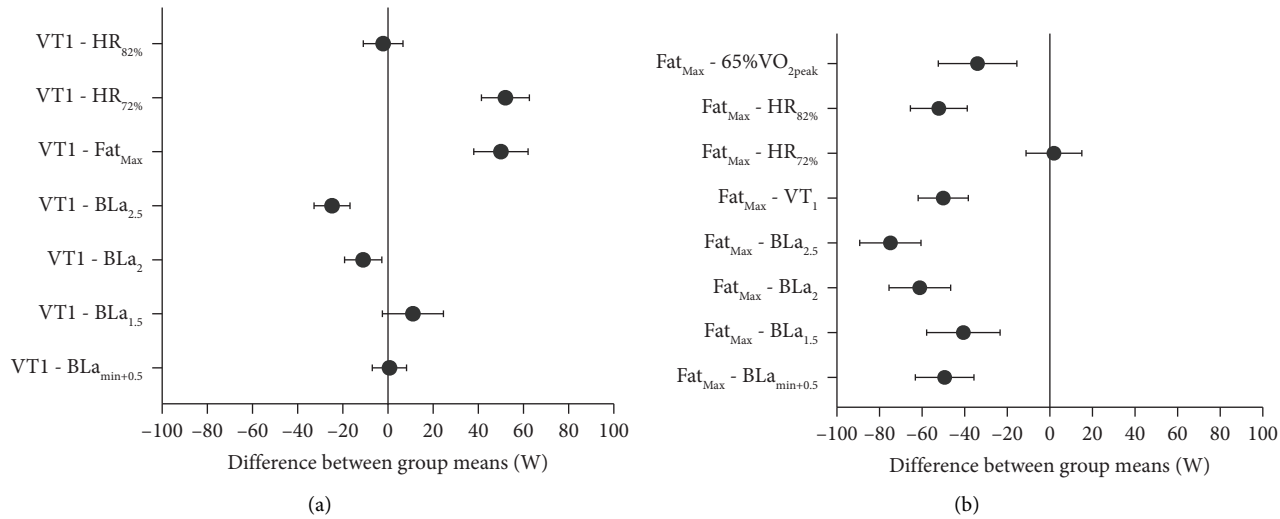

Figure 3: Mean difference in power output (W) between variables with statistical overlay, relative to (a) VT_1 and (b) Fat_{Max} as reference points.

while minimizing carbohydrate consumption during exercise [90].

Blood lactate concentration also offers insights into energy metabolism during exercise [91]. The "P_2" threshold, which indicates the onset of rapid lactate accumulation, suggests a transition to greater carbohydrate utilization and reduced fat oxidation [41, 91]. However, our findings indicate that using P_2 as a proxy for, e.g., Fat_{Max}, is not advisable, as in many instances, $P_{1.5}$ more accurately reflects the threshold below which lipid metabolism is maximized [45]. Moreover, relying on fixed blood lactate concentrations lacks the individualized precision recommended for performance enhancement. This is particularly relevant for blood lactate assessments, where adopting an individualized approach involves simple recalculations rather than changes to the underlying methodology.

4.1. The Dilemma of Individualized vs. Universal Approaches to Z2 Training Prescription. The findings of this study highlight the challenges in defining and prescribing Z2 training intensities using standard methods. The significant interindividual variability observed in Z2 markers underscores the limitations of generalized indicators, such as fixed percentages of heart rate maximum ($\%HR_{max}$), PPO, or standardized blood lactate concentrations. Traditional benchmarks such as 72%–82% HR_{max} or blood lactate levels between 1.5 and 2.5 mmol/L [92] offer a broad framework, but they do not consistently reflect the true metabolic demands across athletes, explaining heterogeneous training adaptations.

Our results indicate that relying on universal benchmarks can result in training loads that are either insufficiently challenging or overly taxing, which can diminish training effectiveness [13]. This variation poses a fundamental dilemma for exercise physiologists and coaches: should training prescriptions prioritize the simplicity of generalized markers, or the precision of individualized markers that better capture the athlete's unique physiological responses?

Furthermore, we are not yet certain which Z2 boundary metric is optimal for achieving specific training outcomes, as this depends on the targeted training goals. The ideal boundary metric may vary based on factors such as the athlete's training background, specific sports demands, and desired adaptations (e.g., enhancing fat oxidation, maximizing mitochondrial density, or improving metabolic flexibility) [44, 93–95]. Generalized prescriptions based on $\%HR_{max}$ or fixed blood lactate concentrations are popular due to their ease of use, but our data suggest that they do not adequately account for interindividual variability, potentially leading to mismatches between prescribed and actual Z2 boundaries, affecting the desired physiological load.

Adopting an individualized exercise intensity approach is essential to improve training precision and effectiveness [96–99]. Specifically, Z2 intensity can be more accurately determined through individualized physiological markers such as VT1, $BLa_{min+0.5}$, or Fat_{Max}. Fat_{Max}, for instance, allows for precise targeting of intensities that optimize fat oxidation [45], an adaptation crucial for endurance performance [41]. This metric is particularly valuable for athletes aiming to enhance endurance, support glycogen sparing, or improve body composition.

However, each marker comes with trade-offs, and the choice of metric should consider athlete-specific factors such as their metabolic efficiency, current fitness level, competitive goals, and access to specialized equipment [100, 101]. VT_1 or markers with low CV may provide more accurate and consistent Z2 training intensities across athletes, allowing coaches to better target desired adaptations such as improved mitochondrial function or enhanced fat metabolism [44]. Although generalized markers are convenient and provide useful guidelines where direct measurements are not feasible, individualized markers—by aligning training more closely with each athlete's unique physiological profile—can optimize training outcomes more effectively [102].

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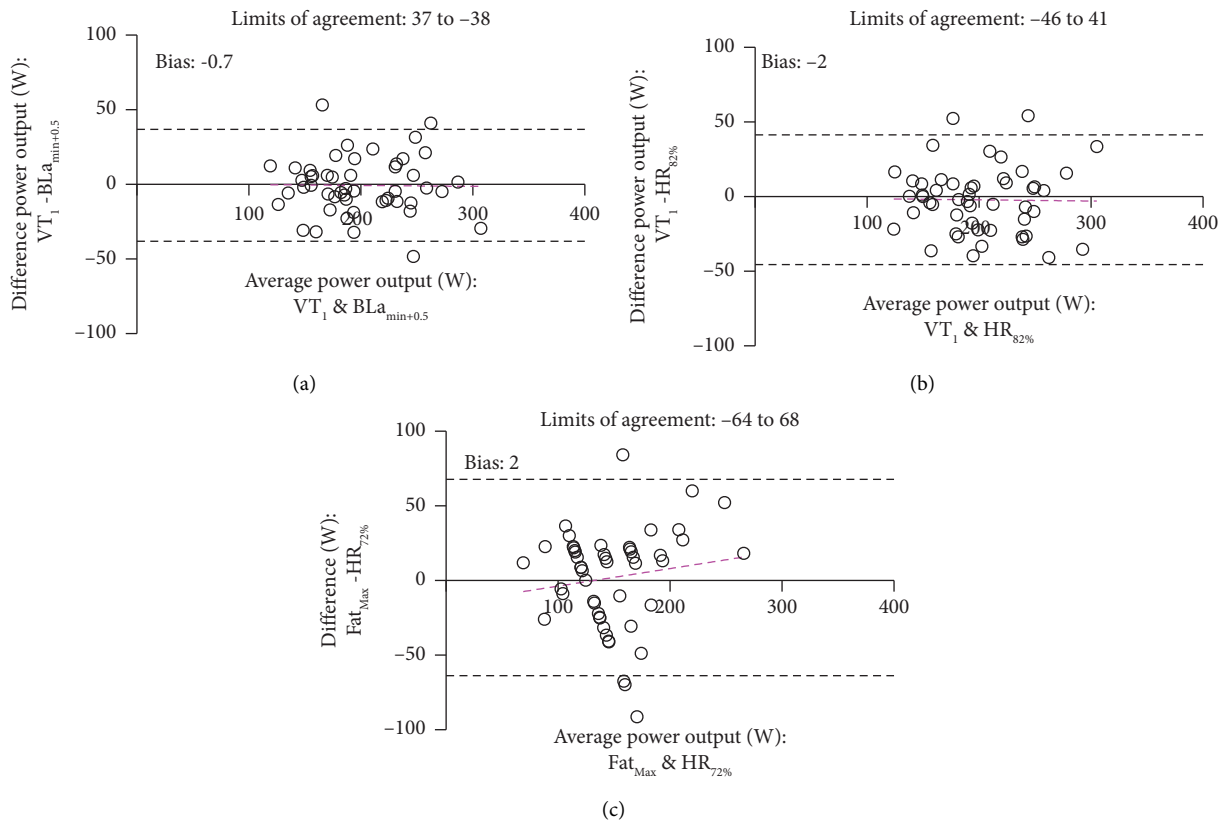

Figure 4: Bland–Altman plots of differences in power output between variables with statistical overlay: (a) VT_1 vs. $BLa_{min+0.5}$, (b) VT_1 vs. $HR_{82\%}$, and (c) Fat_{Max} vs. $HR_{72\%}$.

In summary, individualized prescriptions for Z2 training represent a significant advancement in tailoring low-intensity training for endurance adaptation. While generalized approaches are practical, individualized markers such as VT_1 or Fat_{Max} provide a more precise framework for achieving targeted metabolic adaptations, creating a balanced approach to enhancing training efficacy and athlete performance across diverse training goals [102].

4.2. Training Intensity Distributions. A similar dilemma applies not only to the prescription of training through defined zones but also to the description of training intensity distributions (TIDs) [103]. Meta-analyses of TIDs across endurance disciplines reveal a wide range of definitions for low-intensity Z2 training [2, 92]. Our data suggest that these differing definitions lead to substantial variation in actual training intensity and metabolic load, which in turn affects the comparability across studies, methodologies, and disciplines [103].

Our findings reveal that the relatively small limits of agreement and minimal bias between VT_1 and $BLa_{min+0.5}$ (Figure 4(a)) suggest that these two individually determined threshold markers are well-aligned and thus comparable for defining Z2. While other Z2 demarcations exhibit some degree of overlap, the Bland–Altman analysis highlights increased bias and wider limits of agreement in the comparison of VT_1 to $HR_{82\%}$. Importantly, VT_1 and $BLa_{min+0.5}$ represent demarcations based on physiological markers, whereas $HR_{82\%}$ coincides with these thresholds without being grounded in a direct metabolic response to incremental exercise [104]. In the case of Fat_{Max} and $HR_{72\%}$, although no mean differences were detected, the large limits of agreement observed indicate that these two markers are not interchangeable as Z2 demarcations due to their considerable variability.

For other threshold demarcations, however, intra-individual differences become more pronounced, resulting in distinct metabolic challenges depending on the chosen threshold definition. This raises a critical question about the "true" nature of Z2: although often equated with low-intensity exercise, Z2 can represent quite different physiological demands depending on the variable used to define it [6]. This variability suggests that Z2 may not be a single, universally applicable intensity domain, but rather a concept that encompasses different metabolic states and adaptations depending on the specific marker [6].

4.3. Strengths, Limitations, and Future Directions. A notable strength of this study is the use of a cohort of experienced amateur cyclists, providing insight into the variability of physiological markers even among trained individuals. The inclusion of both ramp and incremental step testing protocols also allowed for a comprehensive evaluation of multiple Z2 intensity markers.

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However, several limitations should be acknowledged. The cross-sectional nature of this study prevented us from evaluating the long-term effects of individualized versus generalized Z2 on training. Additionally, the method used to calculate Fat_{Max} may have introduced variability that influenced the precision of this marker. We consider it a strength of our study that both a ramp protocol and a step protocol were performed. However, when comparing metrics across both protocols, this bears the potential to skew the analysis. Furthermore, our study included only an analysis of cycling and was unable to consider differences in modality [105].

Future studies should include a broader range of endurance athletes as well as modalities and consider additional metrics such as functional threshold power (FTP) and muscle oxygen saturation or employ longitudinal designs to assess the impact of individualized prescriptions over time. Additionally, the possible differences in metabolic response to a single dose of constant Z2 training defined by various demarcations (e.g., Fat_{Max}, VT_1, BLa_2) may provide further insight into this area.

5. Perspective

This study underscores the complexity and challenge of defining Z2 training in a manner that aligns with individual physiological responses, highlighting the limitations of one-size-fits-all prescriptions. Acknowledging the significant interindividual variability across markers such as heart rate, lactate thresholds, oxygen uptake, and power output, there is a compelling case for moving toward more personalized training frameworks. While generalized markers offer convenience and can serve as useful starting points, our findings emphasize that tailored prescriptions based on individualized markers—such as VT1, $BLa_{min+0.5}$, or Fat_{Max}—may provide a more accurate reflection of each athlete's unique metabolic profile, thereby enhancing the potential for targeted adaptations. Moving forward, the integration of personalized Z2 markers into both research and practical training programs could refine endurance training, enabling athletes to optimize their low-intensity efforts and ultimately their performance outcomes. This perspective encourages further investigation into the nuances of individualized versus standardized Z2 prescriptions, potentially leading to more refined guidelines for practitioners in sports science and coaching.

6. Conclusion

The present data underscore the substantial interindividual variability in physiological responses to Z2 intensity prescribed using fixed markers such as % HR_{max} or blood lactate concentrations, challenging the efficacy of traditional methods and highlighting the need for personalized training approaches. By adopting individualized markers such as VT_1 and Fat_{Max}, athletes can achieve more tailored intensity prescriptions that optimize fat metabolism and aerobic performance. Notably, Z2—often equated with low-intensity exercise—can represent distinct metabolic demands depending on the boundaries used, suggesting that it may not be a single, universally applicable intensity domain. Instead, Z2 may encompass varied metabolic states depending on the threshold marker, reinforcing the value of individualized approaches for effective endurance training.

Nomenclature

BLa	Capillary blood lactate
CI	Confidence intervals
CV	Coefficient of variation
Fat_{Max}	Intensity corresponding to peak fat oxidation
FTP	Functional threshold power
HR	Heart rate
HR_{max}	Maximal heart rate
HRR	Heart rate reserve
MAD	Median absolute deviation
PPO_{ramp}	Peak power output during the ramp test
PPO_{step}	Peak power output during the step test
T1	First visit to the laboratory
T2	Second visit to the laboratory
VO_{2peak}	Peak oxygen uptake
VT_1	Ventilatory threshold 1
Z2	Zone 2

Data Availability Statement

The data that support the findings of this study are available on reasonable request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Conflicts of Interest

The authors declare no conflicts of interest.

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